

# KidzKare Insurance

## Data Warehouse – Star Schema Documentation

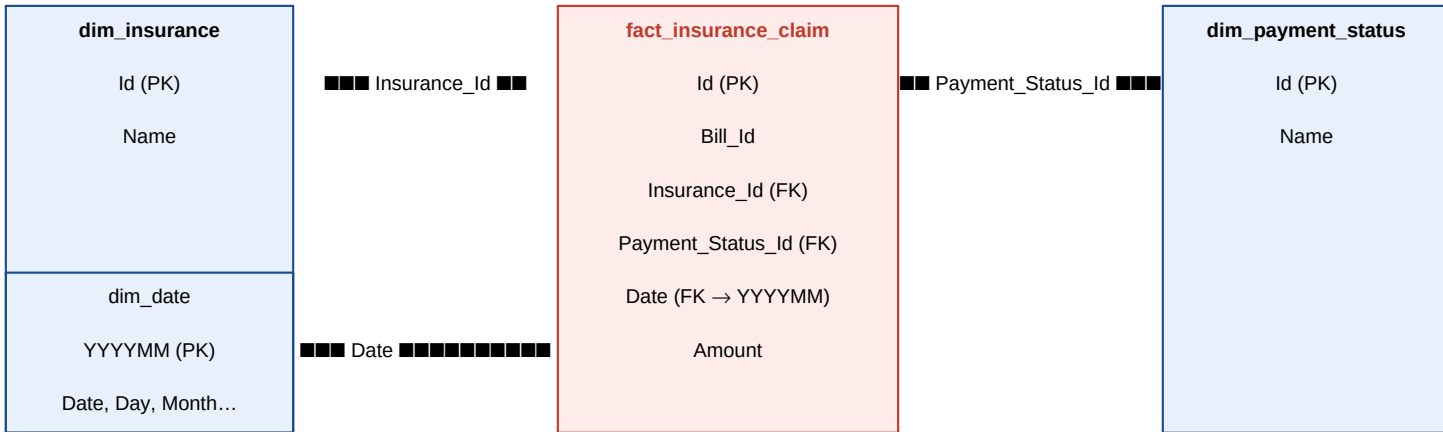
Database: **KidzKare\_Insurance** | Schema: **dbo** | Generated: March 30, 2026

### 1. Overview

This document describes the star schema used in the **KidzKare\_Insurance** data warehouse. The schema is designed to support analytical queries on insurance claim transactions, enabling reporting by insurance provider, payment status, and date dimensions.

The schema follows the **Kimball star schema** pattern: one central fact table surrounded by denormalised dimension tables. All foreign-key relationships use *varchar* surrogate keys.

### 2. Entity-Relationship Overview



**Relationship summary:** fact\_insurance\_claim.Insurance\_Id → dim\_insurance.Id | fact\_insurance\_claim.Payment\_Status\_Id → dim\_payment\_status.Id | fact\_insurance\_claim.Date → dim\_date.YYYYMM

### 3. Relationships

| From Column                              | To Column             | Cardinality | Notes                                               |
|------------------------------------------|-----------------------|-------------|-----------------------------------------------------|
| fact_insurance_claim.Insurance_Id →      | dim_insurance.Id      | Many-to-One | Links each claim to an insurance provider           |
| fact_insurance_claim.Payment_Status_Id → | dim_payment_status.Id | Many-to-One | Links each claim to its payment status              |
| fact_insurance_claim.Date →              | dim_date.YYYYMM       | Many-to-One | Links each claim to a month-year period (YYYYMM for |

## 4. Table Definitions

### 4.1 fact\_insurance\_claim [FACT TABLE]

Central fact table that records each individual insurance claim transaction. Each row represents one claim event tied to a bill, an insurance provider, a payment status, and a reporting month.

| Column            | Data Type    | Nullable | Description                                                                       |
|-------------------|--------------|----------|-----------------------------------------------------------------------------------|
| Id                | VARCHAR(MAX) | YES      | Surrogate primary key uniquely identifying each claim record                      |
| Bill_Id           | VARCHAR(MAX) | YES      | Reference to the source billing system record (bill identifier)                   |
| Insurance_Id      | VARCHAR(MAX) | YES      | Foreign key → dim_insurance.Id; identifies the insurance provider                 |
| Payment_Status_Id | VARCHAR(MAX) | YES      | Foreign key → dim_payment_status.Id; current payment status of the claim          |
| Date              | VARCHAR(MAX) | YES      | Foreign key → dim_date.YYYYMM; reporting period in YYYYMM format (e.g., '202403') |
| Amount            | FLOAT        | YES      | Monetary value of the insurance claim in the local currency                       |

### 4.2 dim\_insurance [DIMENSION TABLE]

Lookup / dimension table that holds the list of insurance providers. Each row represents a unique insurance company or scheme accepted by KidzKare.

| Column | Data Type    | Nullable | Description                                                                           |
|--------|--------------|----------|---------------------------------------------------------------------------------------|
| Id     | VARCHAR(MAX) | YES      | Surrogate primary key for the insurance provider; referenced by fact_insurance_claim. |
| Name   | VARCHAR(MAX) | YES      | Human-readable name of the insurance provider (e.g., 'MedAid', 'CIMAS')               |

### 4.3 dim\_payment\_status [DIMENSION TABLE]

Lookup table enumerating all possible payment states a claim may be in, such as Pending, Paid, Rejected, or Partially Paid.

| Column | Data Type    | Nullable | Description                                                                                         |
|--------|--------------|----------|-----------------------------------------------------------------------------------------------------|
| Id     | VARCHAR(MAX) | YES      | Surrogate primary key for the payment status; referenced by fact_insurance_claim.Payment_Status_Id. |
| Name   | VARCHAR(MAX) | YES      | Descriptive label for the payment status (e.g., 'Paid', 'Pending', 'Rejected')                      |

4.4 dim\_date [DIMENSION TABLE]

Comprehensive date dimension pre-populated with one row per calendar day. The fact table joins on the **YYYYMM** column (month-level granularity), allowing aggregation across all date attributes without additional transformation.

| Column              | Data Type    | Nullable | Description                                                                        |
|---------------------|--------------|----------|------------------------------------------------------------------------------------|
| Date                | VARCHAR(MAX) | YES      | Full calendar date as a string (e.g., '2024-03-15')                                |
| DateS               | BIGINT       | YES      | Date as an integer serial number (days since epoch or YYYYMMDD integer)            |
| Day                 | BIGINT       | YES      | Day-of-month number (1–31)                                                         |
| Weekday             | BIGINT       | YES      | Day-of-week number (1 = Monday ... 7 = Sunday, ISO convention)                     |
| WeekDayName         | VARCHAR(MAX) | YES      | Full English name of the weekday (e.g., 'Monday')                                  |
| IsWeekend           | BIGINT       | YES      | Boolean flag: 1 if the date falls on Saturday or Sunday, else 0                    |
| IsHoliday           | BIGINT       | YES      | Boolean flag: 1 if the date is a public holiday, else 0                            |
| HolidayText         | VARCHAR(MAX) | YES      | Name/description of the public holiday, NULL when IsHoliday = 0                    |
| DayOfYear           | BIGINT       | YES      | Ordinal day within the year (1–366)                                                |
| WeekOfMonth         | BIGINT       | YES      | Week number within the calendar month (1–5)                                        |
| WeekOfYear          | BIGINT       | YES      | ISO week number within the year (1–53)                                             |
| ISOWeekOfYear       | BIGINT       | YES      | Strict ISO 8601 week-of-year number                                                |
| Month               | BIGINT       | YES      | Month number (1–12)                                                                |
| MonthName           | VARCHAR(MAX) | YES      | Full English month name (e.g., 'March')                                            |
| Quarter             | BIGINT       | YES      | Quarter number (1–4)                                                               |
| QuarterName         | VARCHAR(MAX) | YES      | Quarter label (e.g., 'Q1', 'Q2')                                                   |
| Year                | BIGINT       | YES      | Four-digit calendar year (e.g., 2024)                                              |
| YYYYMM              | VARCHAR(MAX) | YES      | <b>JOIN KEY</b>: Six-character month code used as FK target (e.g., '202403'). fact |
| MonthYear           | VARCHAR(MAX) | YES      | Human-readable month-year label (e.g., 'Mar 2024')                                 |
| FirstDayOfMonth     | VARCHAR(MAX) | YES      | Date string of the first day in the current month                                  |
| LastDayOfMonth      | VARCHAR(MAX) | YES      | Date string of the last day in the current month                                   |
| FirstDayOfQuarter   | VARCHAR(MAX) | YES      | Date string of the first day of the current quarter                                |
| LastDayOfQuarter    | VARCHAR(MAX) | YES      | Date string of the last day of the current quarter                                 |
| FirstDayOfYear      | VARCHAR(MAX) | YES      | Date string of 1 January of the current year                                       |
| LastDayOfYear       | VARCHAR(MAX) | YES      | Date string of 31 December of the current year                                     |
| FirstDayOfNextMonth | VARCHAR(MAX) | YES      | Date string of the first day of the following month                                |
| FirstDayOfNextYear  | VARCHAR(MAX) | YES      | Date string of 1 January of the following year                                     |

| Column    | Data Type    | Nullable | Description                                                     |
|-----------|--------------|----------|-----------------------------------------------------------------|
| ValidFrom | VARCHAR(MAX) | YES      | SCD validity start date (slowly changing dimension support)     |
| ValidTo   | VARCHAR(MAX) | YES      | SCD validity end date (NULL indicates the currently active row) |

## 5. Query Guidance for SQL Agent (RAG Context)

The following rules and patterns should be followed when generating SQL queries against this schema:

|                                    |                                                                                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date filtering:</b>             | Always join fact_insurance_claim on fact_insurance_claim.Date = dim_date.YYYYMM. The Date column stores values in YYYYMM string format (e.g., '202403'). Do not treat it as a full date or timestamp. |
| <b>Insurance filtering:</b>        | To filter by insurance provider name, join on fact_insurance_claim.Insurance_Id = dim_insurance.Id and apply a WHERE clause on dim_insurance.Name.                                                    |
| <b>Payment status filtering:</b>   | To filter by claim status (e.g., only 'Paid' claims), join on fact_insurance_claim.Payment_Status_Id = dim_payment_status.Id and filter on dim_payment_status.Name.                                   |
| <b>Aggregations:</b>               | Use SUM(Amount) for total claim values, COUNT(Id) for number of claims. Group by any dim_date attribute (Month, Quarter, Year, MonthName, etc.) for time-series analysis.                             |
| <b>Nullable columns:</b>           | All columns are nullable (no NOT NULL constraints). Always handle NULL values with COALESCE or IS NOT NULL filters where necessary.                                                                   |
| <b>Data types:</b>                 | Most columns are VARCHAR(MAX). Cast to numeric types (CAST/CONVERT) when performing arithmetic on columns like DateS, Day, Month, Year, etc.                                                          |
| <b>Weekend / holiday analysis:</b> | Use dim_date.IsWeekend = 1 or dim_date.IsHoliday = 1 to segment claims that occurred on non-business days.                                                                                            |

### Example Query – Monthly Claims Summary by Insurance Provider

```
SELECT
    d.MonthYear,
    i.Name           AS InsuranceName,
    ps.Name          AS PaymentStatus,
    COUNT(f.Id)      AS TotalClaims,
    SUM(f.Amount)    AS TotalAmount
FROM dbo.fact_insurance_claim f
JOIN dbo.dim_date           d ON f.Date           = d.YYYYMM
JOIN dbo.dim_insurance      i ON f.Insurance_Id    = i.Id
JOIN dbo.dim_payment_status ps ON f.Payment_Status_Id = ps.Id
WHERE d.Year = 2024
GROUP BY d.MonthYear, i.Name, ps.Name
ORDER BY d.MonthYear, i.Name;
```

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This document is intended for use as RAG context for an AI SQL Agent operating against the **KidzKare\_Insurance** database. All table and column names should be qualified with the **dbo** schema prefix when constructing queries (e.g., **dbo.fact\_insurance\_claim**).